3CS524CC26 - FULL STACK DEVELOPMENT

PRACTICAL – 5

Demonstrate the Implementation of Bootstrap Online and Offline Mode Through HTML Webpages

Bootstrap 5.3.3

Course Code	3CS524CC26 - Full Stack Development
Roll No.	24BTM026
Division	F2
Practical	5

1. Aim

To demonstrate the implementation of Bootstrap in HTML webpages using both online mode through a CDN and offline mode through locally stored Bootstrap CSS and JavaScript files.

2. Objectives

• Understand Bootstrap and its responsive, mobile-first approach.

• Implement Bootstrap using CDN links in an online HTML webpage.

• Implement Bootstrap using local Bootstrap files in an offline HTML webpage.

• Demonstrate buttons, cards, forms, alerts, badges, navbar and the responsive grid system.

• Compare online and offline Bootstrap implementation.

3. Software / Requirements

• HTML5-compatible browser.

• VS Code or another code editor.

• Internet connection for online/CDN mode.

• Bootstrap 5.3.3 CSS and JavaScript files for offline mode.

4. Theory – Bootstrap

Bootstrap is a front-end framework for creating responsive and mobile-first websites. It provides predefined CSS classes, a 12-column responsive grid, utility classes and JavaScript-based components. The supplied local Bootstrap files are version 5.3.3.

5. Online Mode

In online mode, Bootstrap is loaded from a remote CDN. The submitted online page uses Bootstrap 5.3.3 CSS and the Bootstrap bundle JavaScript from jsDelivr.

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
      rel="stylesheet">

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
```

6. Online Demo – Code

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Bootstrap Online Demo</title>

    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
          rel="stylesheet">
</head>

<body>
<div class="container py-5">
    <h1 class="text-center mb-4">Bootstrap Online Demo</h1>

    <h2>Buttons</h2>
    <button class="btn btn-primary me-2">Primary</button>
    <button class="btn btn-secondary">Secondary</button>

    <h2 class="mt-4">Alert</h2>
    <div class="alert alert-info" role="alert">
        This is an info alert—check it out!
    </div>

    <h2>Card</h2>
    <div class="row mt-3">
        <div class="col-md-4">
            <div class="card">
                <div class="card-body">
                    <h5 class="card-title">Card Title</h5>
                    <p class="card-text">
                        Sample Bootstrap card component.
                    </p>
                    <a href="#" class="btn btn-primary">Button</a>
```

```
                </div>
            </div>
        </div>
    </div>

    <h2 class="mt-4">Form Example</h2>
    <form>
        <div class="mb-3">
            <label class="form-label">Email address</label>
            <input type="email" class="form-control">
        </div>
        <div class="mb-3">
            <label class="form-label">Password</label>
            <input type="password" class="form-control">
        </div>
        <button type="submit" class="btn btn-primary">Submit</button>
    </form>
</div>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

7. Offline Mode

In offline mode, Bootstrap files are stored in the project directory and linked with relative paths. This removes the need to download Bootstrap from a CDN while the webpage is running.

```
<link href="bootstrap.min.css" rel="stylesheet">

<script src="bootstrap.bundle.min.js"></script>
```

8. Offline Components – Code

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Bootstrap Components Demo</title>
    <link href="bootstrap.min.css" rel="stylesheet">
</head>

<body>
<nav class="navbar navbar-expand-lg navbar-dark bg-primary mb-4">
    <div class="container">
        <a class="navbar-brand" href="#">Bootstrap Demo</a>
        <button class="navbar-toggler" type="button"
                data-bs-toggle="collapse"
                data-bs-target="#navbarNav">
            <span class="navbar-toggler-icon"></span>
        </button>

        <div class="collapse navbar-collapse" id="navbarNav">
            <ul class="navbar-nav">
                <li class="nav-item">
                    <a class="nav-link active" href="#">Home</a>
                </li>
                <li class="nav-item">
                    <a class="nav-link" href="#">Features</a>
                </li>
                <li class="nav-item">
                    <a class="nav-link" href="#">Pricing</a>
                </li>
            </ul>
        </div>
    </div>
</nav>

<div class="container">
    <h2>Buttons</h2>
    <button class="btn btn-primary">Primary</button>
    <button class="btn btn-secondary">Secondary</button>
    <button class="btn btn-success">Success</button>
    <button class="btn btn-danger">Danger</button>

    <h2 class="mt-4">Cards</h2>
    <div class="row">
        <div class="col-md-4 mb-3">
            <div class="card">
                <div class="card-body">
                    <h5 class="card-title">Card Title</h5>
                    <p class="card-text">Sample card component.</p>
                    <a href="#" class="btn btn-primary">Button</a>
                </div>
            </div>
        </div>
    </div>

    <h2 class="mt-4">Forms</h2>
    <form>
        <div class="mb-3">
            <label class="form-label">Email address</label>
```

```
            <input type="email" class="form-control">
        </div>
        <button type="submit" class="btn btn-primary">Submit</button>
    </form>

    <h2 class="mt-4">Alerts</h2>
    <div class="alert alert-primary">Primary alert</div>
    <div class="alert alert-success">Success alert</div>
    <div class="alert alert-danger">Danger alert</div>
    <div class="alert alert-warning">Warning alert</div>

    <h2>Badges</h2>
    <h3>Example heading
        <span class="badge bg-secondary">New</span>
    </h3>
</div>

<script src="bootstrap.bundle.min.js"></script>
</body>
</html>
```

9. Bootstrap Grid System – Code

The grid demonstration uses Bootstrap's 12-column responsive system and changes column widths at different breakpoints.

```
<div class="container mt-5">
    <h1>Bootstrap Grid System Demo</h1>

    <div class="row mb-4">
        <div class="col-12 col-md-6 col-lg-4 box">
            <h5>.col-12 .col-md-6 .col-lg-4</h5>
            <p>Full width on extra small, half on medium,
               one-third on large.</p>
        </div>

        <div class="col-12 col-md-6 col-lg-4 box">
            <h5>.col-12 .col-md-6 .col-lg-4</h5>
            <p>Responsive Bootstrap column.</p>
        </div>

        <div class="col-12 col-md-6 col-lg-4 box">
            <h5>.col-12 .col-md-6 .col-lg-4</h5>
            <p>Responsive Bootstrap column.</p>
        </div>
    </div>

    <div class="row">
        <div class="col-2 box bg-secondary text-white">.col-2</div>
        <div class="col-8 box bg-info text-white">.col-8</div>
        <div class="col-2 box bg-dark text-white">.col-2</div>
    </div>
</div>
```

10. Student Registration Form – Code

```
<div class="container">
    <div class="form-container">
        <h2 class="form-title">Student Registration Form</h2>

        <form id="studentForm" novalidate>
            <div class="row g-3">
                <div class="col-md-6">
                    <label for="rollNumber" class="form-label">
                        Roll Number
                    </label>
                    <input type="text" class="form-control"
                           id="rollNumber" required>
                    <div class="invalid-feedback">
                        Please provide a roll number.
                    </div>
                </div>

                <div class="col-md-6">
                    <label for="firstName" class="form-label">
                        First Name
                    </label>
                    <input type="text" class="form-control"
                           id="firstName" required>
                    <div class="invalid-feedback">
                        Please provide your first name.
                    </div>
                </div>
            </div>

            <div class="mb-3">
                <label class="form-label">Mobile Number</label>
                <input type="tel" class="form-control"
                       pattern="[0-9]{10}" required>
            </div>

            <div class="mb-3">
                <label class="form-label">Email ID</label>
                <input type="email" class="form-control" required>
            </div>
```

```
            <div class="mb-3">
                <label class="form-label">Address</label>
                <textarea class="form-control" rows="3" required></textarea>
            </div>

            <div class="d-grid">
                <button type="submit" class="btn btn-primary">
                    Register
                </button>
                <button type="reset" class="btn btn-outline-secondary">
                    Reset
                </button>
            </div>
        </form>
    </div>
</div>
```

11. Registration Form – Validation JavaScript

```
(function () {
    'use strict'

    const form = document.getElementById('studentForm')

    form.addEventListener('submit', function (event) {
        event.preventDefault()

        if (form.checkValidity()) {
            alert('Registration successful! Redirecting to home page...')
            window.location.href = 'index.html'
        } else {
            form.classList.add('was-validated')
        }
    })
})()
```

12. Bootstrap Components Demonstrated

• Navbar – responsive navigation.

• Buttons – primary, secondary, success, danger and other variants.

• Cards – structured content using card classes.

• Forms – labels, inputs and validation feedback.

• Alerts – contextual messages.

• Badges – small status indicators.

• Grid – responsive 12-column layout.

• Utilities – spacing, alignment, typography and colors.

13. Online vs Offline Comparison

Feature	Online Mode	Offline Mode
Bootstrap source	CDN	Local project files
Internet for Bootstrap	Required	Not required
CSS	Remote Bootstrap CSS	bootstrap.min.css
JavaScript	Remote bundle	bootstrap.bundle.min.js
Setup	Very quick	Requires local files
Best use	Quick web demos	Labs and no-network environments

14. Procedure

1. Create an HTML5 webpage.

2. For online mode, add Bootstrap 5.3.3 CSS and JS CDN links.

3. Add Bootstrap components such as buttons, alerts, cards and forms.

4. Open the page with Internet access and verify the components.

5. For offline mode, place bootstrap.min.css and bootstrap.bundle.min.js in the project folder.

6. Reference the local files from the HTML page.

7. Open the page without Internet and verify that Bootstrap still works.

8. Resize the browser to verify responsive behavior.

15. Important Bootstrap Classes

Class	Purpose
container	Responsive content container
row	Grid row
col-md-6	Six of twelve columns at medium breakpoint
col-lg-4	Four of twelve columns at large breakpoint
btn btn-primary	Primary button
card / card-body	Card component
form-control	Styled form input
form-label	Form label
alert alert-info	Information alert
badge bg-secondary	Badge
navbar	Navigation bar
d-grid	Grid display utility
mb-3 / mt-3	Spacing utilities
text-center	Center-align text

16. Result

Bootstrap 5.3.3 was successfully implemented in both online and offline modes. The practical demonstrates CDN-based integration, local Bootstrap integration, responsive grid layouts, navigation, buttons, cards, alerts, badges, forms and validation.

17. Conclusion

Bootstrap simplifies responsive web development through predefined CSS classes, a grid system, utilities and components. Online mode provides a quick CDN-based setup, while offline mode makes Bootstrap available from local project files. Both approaches were demonstrated using the submitted HTML pages.